ReviewRadar :

FROM REVIEWS TO PRIORITIES

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The problem

- Hundreds of reviews, no clear starting point.
- Reading customer reviews one by one takes too much time.
- Recurring complaints get lost among positive and unrelated comments.
- Affects small business owners and sellers with no resources to analyse feedback manually.

Who Faces This Problem?

- Online business owners and independent sellers.
- Sellers who receive customer feedback but lack the time or resources to analyse it manually.

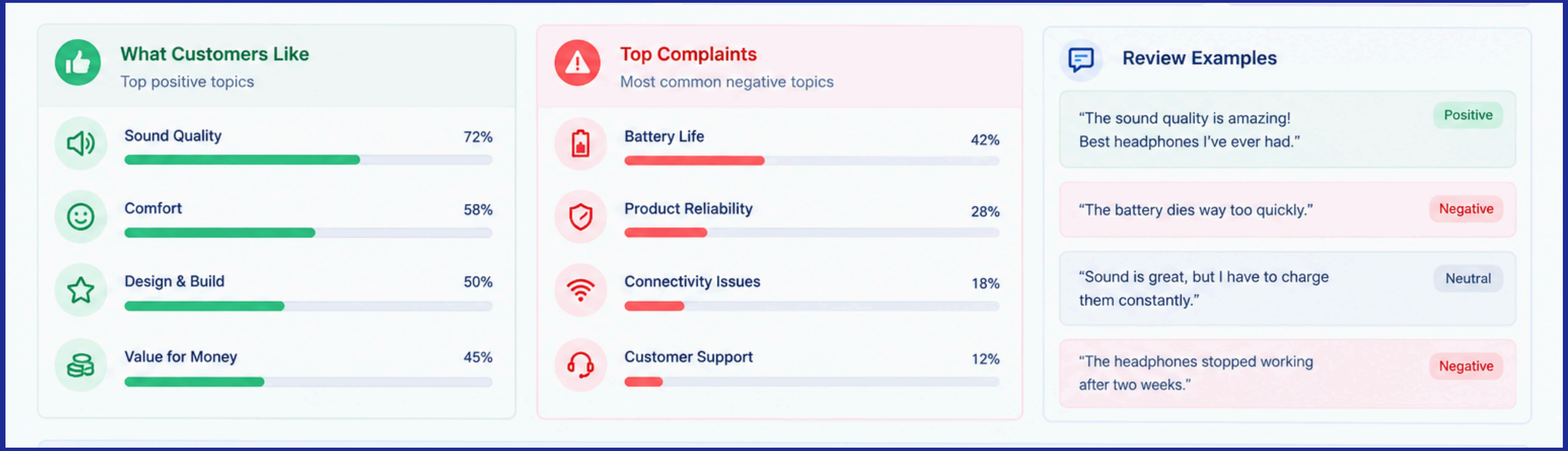
THE OPPORTUNITY

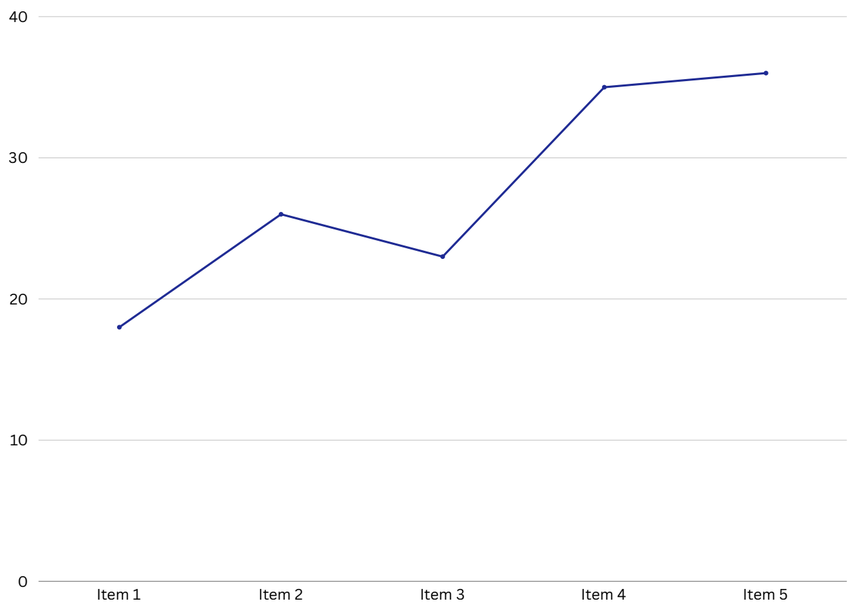
Repeated complaints show what to fix first. Customer feedback can directly improve product quality and satisfaction.

Proposed Solution and USP

ReviewRadar is a web based tool where sellers can upload a CSV file containing customer reviews. It analyses the reviews, identifies their sentiment and groups similar complaints into topics such as battery life, product quality, or delivery delays.
The dashboard displays frequently reported issues, their supporting customer reviews and a priority score to help sellers decide which problems to investigate first.

USP: Instead of just analysing reviews, ReviewRadar helps sellers prioritise recurring customer complaints so they know what needs attention first.





Performance by Product

Use of AI/ML

SENTIMENT ANALYSIS

Uses supervised machine learning to classify customer reviews as positive, negative, or neutral. The model learns from previously labelled reviews and predicts the sentiment of new reviews, helping sellers understand overall customer satisfaction.

NATURAL LANGUAGE PROCESSING (NLP)

Processes and analyzes customer feedback using text processing and similarity techniques. It identifies reviews with similar meanings, even when customers use different words to describe the same problem.

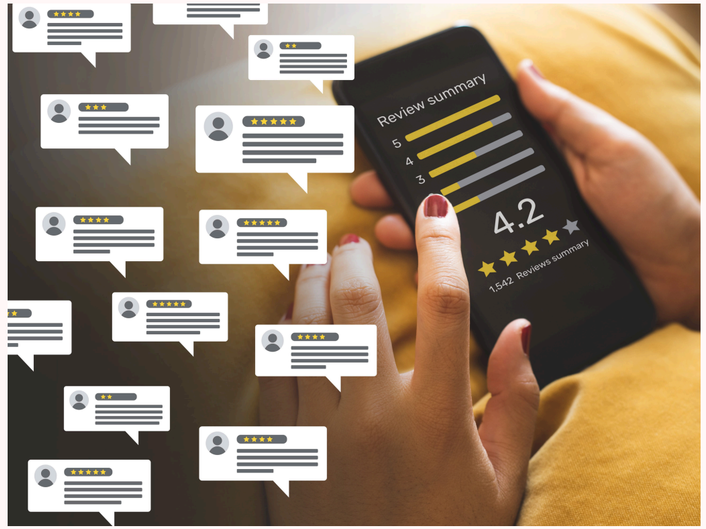
COMPLAINT CATEGORIZATION

Groups similar reviews into meaningful categories, such as battery life, product quality, or delivery delays. This helps sellers identify recurring issues and understand which aspects of their products or services need improvement.

PRIORITY SCORING

Uses a transparent, rule-based formula that considers factors such as complaint frequency and customer ratings to assign priority scores to identified issues. This helps sellers focus on the most frequently reported and significant problems first.

Workflow of customer insights



1. **Upload CSV File**
   - Add your customer reviews in CSV format.
2. **Analyse Sentiment**
   - See the overall sentiment distribution of your reviews.
3. **Group Similar Complaints**
   - Discover recurring complaint categories using NLP.
4. **Calculate Priority**
   - Each issue gets a priority score based on frequency and ratings.
5. **View Original Reviews**
   - Read the actual customer reviews behind each finding.
6. **Take Action**
   - Compare issues and focus on what needs attention first.

What Data Will Be Needed?

The project will primarily use customer review data, including:

- Review text.
- Star rating.
- Product name or category, if available.
- Review date

The final product would look like a web-based dashboard where sellers upload customer reviews and view sentiment analysis, recurring complaints, original feedback and priority scores to decide what to improve first.

Challenges and Uncertainties

- Customer reviews may contain sarcasm, unclear wording, or mixed opinions.
- Similar complaints may be expressed in different ways.
- The sentiment model may incorrectly classify some reviews.
- Deciding how much importance to give complaint frequency and customer ratings will require testing.

Similar Existing Solutions

1. Chattermill

An AI-powered customer feedback platform that analyses sentiments and helps businesses understand customer feedback.

Website: https://chattermill.com/

2. unitQ

A platform that uses AI to analyse customer feedback and identify product issues and emerging trends.

Website: https://www.unitq.com/product-overview/

How our product is different?

Our product focuses on helping small online sellers prioritise recurring customer complaints through a simple dashboard, transparent priority scores, and actual customer reviews supporting each finding, making it easier to decide what to improve first.

TECH STACK

Component	Technology	Purpose
Frontend	HTML, CSS, JavaScript	Build the dashboard
Backend	Python + Flask	Handle uploaded reviews and process data
Data Processing	Pandas, NumPy	Clean and analyse review data
Machine Learning	Scikit-learn	Train the sentiment analysis model
NLP	NLTK, TF-IDF	Process review text and identify similar complaints
Visualisation	Matplotlib, Plotly	Display sentiment charts and complaint statistics
Data Storage	CSV files initially	Store and process customer reviews
Deployment	Vercel	Host the web application

References

1. Chattermill: https://chattermill.com/
2. Scaling Feedback Programs: https://getthematic.com/insights/review-analysis
3. DEMO: Sentiment Analysis of Product Reviews for a Retail Company
   https://youtu.be/ExzatCJUbio?si=dDSN_bshgna7dpQ4